

Large-scale Models and Robotics: A Comprehensive Review

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Abstract

Despite the rapid advancement in integrating large-scale models (LSMs) into robotics, a comprehensive synthesis of this field remains absent. This study addresses this gap by conducting a bibliometric analysis and critical review, focusing on the technical mechanisms and application frontiers of LSM-robotics integration. The research covers five pivotal areas where LSMs drive notable progress in robotics: multimodal perception, semantic planning, human-robot interaction, cross-domain generalization, and efficient system deployment—with tangible improvements evidenced by reduced task planning errors, accelerated sensor fusion, and enhanced safety validation reliability. Current trends indicate that LSMs are transitioning robotic systems from pre-programmed routines to adaptive, intelligent platforms, though challenges such as real-time computation bottlenecks and environmental robustness gaps persist. By synthesizing a decade of research, this review provides a structured framework to guide the real-world implementation of LSM-enhanced robotic systems, which is crucial for advancing the field toward more autonomous, reliable, and human-centric intelligent robotics.

1 Introduction

The field of robotics is undergoing a transformative shift from pre-programmed routines toward adaptive, intelligent systems capable of understanding and interacting with complex, unstructured environments. This evolution is primarily driven by advancements in artificial intelligence (AI), particularly the development of large-scale models (LSMs) trained on massive datasets [1]. Competences in natural language processing, multimodal understanding, and content generation—demonstrated by key architectures such as large language models (LLMs) [2], vision-language models (VLMs) [3], and generative models like diffusion processes [4]—are now being transferred to robotics. This integration is paving the way for a new generation of robots capable of semantic reasoning, cross-task generalization, and intuitive human-robot collaboration [5].

The convergence of LSMs with robotics enhances several core robotic capabilities, including perception, planning, control, and interaction. For example, LLMs can interpret high-level natural language commands and decompose them into structured task plans [6], while VLMs allow robots to query and interpret visual scenes semantically [3]. Furthermore, generative models can create synthetic training data to bridge the simulation-to-reality gap [7] or even generate potential motion trajectories [8]. This represents a paradigm shift beyond traditional robotics, which often struggles with unforeseen environmental variability and abstract commands without extensive, task-specific programming.

Despite rapid progress, existing reviews frequently focus on a singular category of LSM, such as LLMs, thereby overlooking

the broader synergistic potential of diverse LSMs within integrated robotic systems [9]. A comprehensive synthesis is urgently needed to explore how these different models collectively address fundamental challenges such as multimodal sensor fusion, long-horizon task planning, real-time safe control, and robust cross-domain generalization. Moreover, there remains a notable gap in the systematic comparison of empirical performance metrics—including accuracy, latency, and robustness—across different LSM-based approaches. Such comparisons are critical for guiding real-world deployment and establishing reliable benchmarks [10].

To address these gaps, this review provides a systematic analysis of the integration of large-scale models into robotics, focusing on technical mechanisms and application frontiers. It categorizes and analyzes advancements across five critical domains: (1) multimodal perception, (2) semantic task planning and control, (3) human-robot interaction, (4) cross-domain generalization, and (5) efficient system integration. For each domain, it highlights representative methodologies, presents comparative performance analyses where available, and discusses persistent challenges such as real-time computation, safety assurance, and data efficiency. By synthesizing a wide body of research, this review aims to provide a clear roadmap of the current state-of-the-art, identify key hurdles, and outline promising directions for future research in LSM-powered robotics.

The remainder of this paper is organized as follows: Section 2 systematically categorizes and analyzes advancements of LSMs in robotics across five critical application domains—multimodal perception, semantic task planning and control,

human-robot interaction, cross-domain generalization, and efficient system integration—with quantitative performance comparisons for representative approaches. Section 3 synthesizes persistent challenges in technical (e.g., real-time performance, robustness), operational (e.g., standardization), and ethical (e.g., safety governance) dimensions, while outlining promising future research frontiers. Finally, Section 4 concludes the paper by summarizing the key findings and highlighting the pathway toward widespread real-world deployment of LSM-enhanced robotic systems.

2 Five Major Application Domains of LSMs in Robotics

2.1 Perception Enhancement: Multimodal Scene Understanding

The enhancement of perceptual capabilities is among the most significant contributions of Large-scale Models (LSMs) to robotics. Moving beyond traditional computer vision techniques, LSMs enable robots to achieve a more semantic, contextual, and robust understanding of their environment by fusing and interpreting data from multiple sensors (e.g., vision, depth, tactile) through a unified model architecture [11].

A pivotal advancement is the application of Vision-Language Models (VLMs) for semantic scene understanding. Models like CLIP [3] learn a joint embedding space for images and text, allowing robots to interpret scenes based on natural language queries. This capability is transformative for tasks such as industrial bin-picking, where robots must identify and grasp specific objects from cluttered backgrounds based on semantic instructions (e.g., "pick up the red gear") rather than pre-defined templates. This approach markedly reduces misidentification rates in variable lighting conditions compared to traditional feature-matching methods [12]. Beyond 2D vision, neural rendering techniques, particularly Neural Radiance Fields (NeRFs), have emerged as a powerful tool for 3D scene reconstruction. By synthesizing high-fidelity, novel views of a scene from sparse 2D images, NeRFs provide robots with a dense 3D environmental model, greatly enhancing navigation and manipulation planning in complex environments [13].

Generative models, especially diffusion models, are increasingly being leveraged to augment robotic perception.

They are applied to generate realistic synthetic training data, mitigating the data scarcity problem for rare or hard-to-capture scenarios [14]. Moreover, they show promise in processing and completing (i.e., filling missing parts of) high-dimensional sensory data—for instance, inferring full tactile signals from partial inputs or enhancing low-resolution images in real-time [15].

Several key technical innovations underpin these improvements:

Cross-modal attention mechanisms allow for efficient fusion of heterogeneous sensory data (e.g., LiDAR point clouds with camera images), significantly reducing sensor fusion latency and improving coherence in autonomous navigation systems compared to early fusion or decision-level fusion strategies [16].

Self-supervised learning paradigms, often built upon contrastive learning frameworks, drastically reduce the dependency on vast amounts of manually annotated data. For instance, models can learn effective visual representations from unlabeled video sequences, achieving high accuracy in defect detection or anomaly identification without exhaustive supervision [17].

Yet, several hurdles persist in deploying LSM-enhanced perception systems:

Environmental Sensitivity: Performance can degrade under significant environmental variations, such as extreme lighting changes (>1000 lux) or adverse weather conditions. Although VLMs offer improved robustness, this remains an active area of research [12].

Sensor Asynchrony: Temporal misalignment between heterogeneous sensors (e.g., rolling shutter cameras and high-frequency LiDAR) can introduce errors in dynamic scenes. Solving this requires sophisticated temporal calibration and fusion algorithms beyond standard architectures [18].

Data and Computational Demands: Training and fine-tuning models like NeRFs often require thousands of annotated samples (e.g., >10,000 CT scans for medical NeRF applications [13]), and their inference latency can be prohibitive for real-time control loops, necessitating further work on model compression and efficient inference.

Table 1 quantitatively compares the performance characteristics of prominent LSM-based perception approaches across critical operational parameters, highlighting the inherent trade-offs.

Table 1. Performance Comparison of Multimodal Perception Models in Robotic Applications

No.	Model & Scenario	Key Metric Improvement	Latency (ms)	Data Requirements	Environmental Sensitivity
1	CLIP-VLM (Industrial picking)	↑ Recognition accuracy in clutter	120	>10k annotated images	Sensitive to extreme lighting
2	NeRF (3D Surgical mapping)	<5mm reconstruction error	200	>10k CT scans	Sensitive to tissue deformation
3	Diffusion-Tactile (Prosthetics)	↑ Texture discrimination accuracy	85	>5k tactile signals	Varies with pressure fluctuations
4	Cross-Modal Attention (Navigation)	↓ Fusion latency	65	>50k multimodal sensor samples	Errors from sensor temporal misalignment
5	Diffusion Data Gen. (Sim2Real)	↑ Sim-to-real alignment	Offline	Unlabeled field images	Affected by domain shift

2.2 Planning and Control: Semantic-to-Action Translation

A fundamental breakthrough enabled by Large-scale Models (LSMs) is their capacity to bridge the gap between high-level task abstraction and low-level motor control. This is achieved through sophisticated hierarchical reasoning architectures that interpret ambiguous commands, formulate multi-step plans, and generate executable actions, thereby markedly enhancing the autonomy of robotic systems [19].

A prominent application is the use of Large Language Models (LLMs) for semantic task planning. By leveraging their extensive world knowledge and reasoning capabilities, LLMs can decompose complex, natural language instructions into structured sequences of sub-tasks. For instance, in electronics assembly, GPT-4-based planners have been shown to substantially reduce errors by parsing commands like "assemble the microcontroller board" into a correct sequence of over 50 steps (e.g., "pick up resistor R1", "insert into slot J3", "solder the pins"), while utilizing symbolic memory buffers to track tool-state dependencies throughout the process [6]. This capability is crucial for automating complex manufacturing workflows.

For safety-critical applications, such as autonomous driving, the integration of formal control methods with LSMs has emerged as a key research direction. The Control Barrier Function (CBF)-LLM framework exemplifies this approach. It embeds safety constraints derived from CBFs directly into the attention mechanisms of a transformer model. This ensures that the autonomous vehicle's actions are always constrained to physically feasible and safe regions of the state space, dynamically adjusting trajectories based on real-time risk assessments generated from language-based scene descriptions [20, 21]. This methodology has demonstrated a notable reduction in traffic violations and risky maneuvers in simulation and controlled testing.

Real-time performance is paramount for control applications. To address the computational bottlenecks of standard transformers on resource-constrained platforms, hardware-aware efficient architectures have been developed. For instance, models like MobileViT-Transformer employ grouped convolutions and channel pruning techniques to achieve low-latency control for agile platforms like drones, enabling low-latency navigation and obstacle avoidance on embedded GPUs.

Beyond these specific applications, core technical innovations in LSM-driven planning and control include:

- **Latent Action Spaces:** Models learn compact, low-dimensional representations of motion primitives (e.g., via variational autoencoders) within a latent space. This facilitates more efficient reinforcement learning fine-tuning and policy generalization across related tasks.
- **Failure Prediction Modules:** By integrating predictive models, systems can anticipate and avoid failures. For example, modules based on Hamilton-Jacobi reachability analysis have been integrated to provide up to 94.2% collision recall accuracy in autonomous drone swarms, preemptively triggering evasive maneuvers [22].

Despite these advancements, significant challenges remain. The computational demand of vision transformer-based controllers often prevents them from meeting the strict latency requirements (<100 ms) for low-latency control on many edge devices, a common bottleneck highlighted in performance studies of edge AI deployment [23]. Emerging solutions are focusing on neuromorphic co-processors for energy-efficient, spike-based acceleration [24] and federated learning frameworks for distributed knowledge sharing across multi-robot systems [25].

Table 2 quantifies the operational characteristics of prominent LSM-based planning and control approaches, highlighting critical trade-offs.

Table 2 Performance Comparison of Planning and Control Methods in Robotic Applications

No.	Method & Task Type	Error Reduction / Key Metric	Latency (ms)	Hardware Platform	Limitation / Failure Mode
1	GPT-4 Planner (Industrial Assembly)	Substantially reduces sequencing errors	~200	NVIDIA V100 GPU	Increased failure rate in very long sequences (>50 steps)
2	CBF-LLM Framework (Autonomous Driving)	Drastically reduces traffic violations	50 ± 5	NVIDIA Xavier AGX	Requires accurate dynamic model of the ego-vehicle
3	MobileViT-Transformer (Drone Navigation)	Focused on low-latency performance	50 ± 5	NVIDIA Jetson TX2	Task performance dependent on perception input quality
4	Latent Action Spaces (Mobile Manipulation)	Significantly reduces motion learning samples	~75	Intel i7-12800H	Limited by the diversity of motion primitives in the dataset
5	Failure Prediction Module (Motion Planning)	High collision recall accuracy (94.2%)	~40	AMD Ryzen 9 5950X	High computational cost for real-time prediction

2.3 Human-Robot Interaction: Towards Contextual and Safe Collaboration

The integration of Large-scale Models (LSMs) is

fundamentally reshaping human-robot interaction (HRI), moving it beyond pre-defined scripts towards fluid, contextual, and adaptive collaboration. LSMs enable robots to understand nuanced human intentions, generate appropriate responses,

and adhere to safety guidelines. This makes them critical for deployment in human-centric environments such as healthcare, manufacturing, and domestic settings[26].

A primary application is the use of large language models (LLMs) for robust natural language understanding. This functionality is particularly transformative in healthcare robotics, where instructions are often complex and context-dependent. For instance, models like PaLM-2 have demonstrated high accuracy in enabling nursing robots to parse and execute intricate clinical commands, such as "administer the patient's morning medication and then assist them to the common area" by breaking them down into safe and executable action sequences [27].

In industrial collaboration, vision-language models (VLMs) are being fused with haptic feedback to create material-aware robotic systems. For example, CLIP-guided impedance control systems allow robots to dynamically adjust their grip force based on the semantic understanding of an object's material (e.g., "carbon fiber composite" vs. "steel engine part"), substantially increasing efficiency and preventing damage to delicate components in automotive assembly lines [28].

Ensuring safety and ethical compliance in open-ended HRI is a paramount challenge. Constitutional AI techniques, particularly Reinforcement Learning with Human Feedback (RLHF), are being employed to align robot behavior with human values and safety standards. Models fine-tuned with RLHF can reject unsafe or unethical instructions with high reliability (>99%) by cross-referencing proposed actions

against established ethical frameworks like the IEEE 7009-2021 standard for ethical risk assessment [29].

Despite these advancements, significant deployment challenges persist:

- **Linguistic and Cultural Gaps:** Performance disparities exist across languages. For example, comprehension accuracy for Mandarin and Japanese often lags behind English due to imbalances in the pre-training data of most LSMs, hindering global rollout [27].

- **Computational and Energy Constraints:** Aggressive model quantization techniques, necessary to run LSMs on resource-limited platforms (e.g., Raspberry Pi 4), can lead to a severe degradation in functionality and interaction quality, limiting their use in low-power applications [23].

Emerging research is addressing these bottlenecks. Parameter-efficient fine-tuning methods enable significant customization (e.g., adapting to user accents or specific terminologies) with minimal additional parameters (often under 100KB), making personalized interactions feasible on edge devices. Industrial implementations are already demonstrating the practicality of these solutions, achieving response times of 800ms for voice-controlled welding systems [30] and high instrument handover success rates in clinical trials with the da Vinci surgical platform [28].

Table 3 benchmarks key HRI systems across critical performance dimensions, revealing trade-offs between capability, safety, and deployability.

Table 3. Performance Comparison of Selected Human-Robot Interaction Systems

No.	System & Interaction Type	Accuracy/Improvement	Latency (ms)	Safety Reliability	Deployment Limitation
1	PaLM-2 Nursing Robot (Clinical command parsing)	High accuracy in parsing complex commands	~800	N/A	Performance gap for non-English languages
2	VLM-Haptic System (Material-aware grasping)	Significant efficiency gain, reduced damage	N/A	N/A	Functionality limited on low-power devices post-quantization
3	Constitutional AI (RLHF) (Instruction compliance)	>99% unsafe instruction rejection rate	N/A	IEEE 7009-2021 compliant	Requires extensive human feedback for tuning
4	Low-Rank Adaptation (LoRA) (Personalized eldercare)	Minimal parameters (<100KB) for customization	~800	N/A	Limited scope for broad customization
5	da Vinci Surgical Platform (Instrument handover)	High success rate in clinical trials	~1200	FDA Class II	High cost of deployment (>\$2M)

Note: "N/A" indicates that the original literature did not report data for this dimension, rather than the system not requiring safety assessment. For medical robots (e.g., nursing robots, surgical platforms), safety compliance with standards such as IEEE 7009-2021 or FDA regulations is mandatory in practical deployment.

2.4 Generalization Learning: Achieving Cross-Domain Robustness

A critical challenge in robotics is overcoming the domain shift between training and deployment environments. Large-scale Models (LSMs) address this fundamental problem by enabling robots to acquire robust, transferable skills that generalize across varying conditions, tasks, and embodiments. This capability is primarily realized through innovative frameworks in meta-learning, self-supervised representation learning, and continual adaptation [31].

Meta-learning, or learning to learn, provides a powerful framework for few-shot adaptation. Model-Agnostic Meta-Learning (MAML) algorithms, when combined with Vision-Language Models (VLMs), allow industrial robots to rapidly acquire new skills with minimal data. For instance, a MAML-VLM system can enable a robot to learn a novel object grasping task with high accuracy using only five demonstrations by fine-tuning on a simulated task distribution generated from textual descriptions. This significantly reduces the data and time required for re-tasking in flexible manufacturing applications [32].

In scenarios where labeled data is scarce, self-supervised learning (SSL) offers a compelling solution. Contrastive learning frameworks like SimCLR, integrated with VLMs, can learn powerful visual representations from vast amounts of unlabeled field data[33]. In agricultural robotics, a SimCLR-VLM system has been shown to achieve high cross-crop recognition accuracy (e.g., generalizing from tomatoes to strawberries) by aligning the latent feature spaces of different crop types through a momentum encoder, without requiring manual labels for each new environment [34].

For long-term autonomy in non-stationary environments, continual learning is essential to avoid catastrophic forgetting—the tendency of a model to overwrite previously learned knowledge when trained on new data. Techniques like Elastic Weight Consolidation (EWC) show promise for mitigating this issue when integrated with large language models (e.g., GPT-4), though empirical studies remain limited. Early pilot studies indicate that EWC strategically penalizes updates to network weights that are identified as important for previous tasks (based on Fisher information), thereby enabling

sequential learning across multiple manufacturing tasks with minimal performance degradation on earlier skills [35, 36].

These techniques collectively enhance the robot's ability to generalize across domains.

However, significant challenges to robust generalization remain:

- **Sim-to-Real Gap:** Discrepancies in physical properties (e.g., dynamics, friction, material deformation) between simulation and reality continue to cause significant performance degradation when transferring manipulation skills [37]. Specifically, for multi-stage tasks (e.g., sequential assembly or multi-step object manipulation), end-to-end visuomotor control systems trained in simulation often fail to maintain stability in real-world environments due to unmodeled physical disturbances. For such tasks, this gap is particularly severe—errors in early simulation steps accumulate and cause instability in real-world execution. James et al. [38] addressed this issue by proposing a transfer learning framework that bridges simulation and reality for multi-stage visuomotor tasks, achieving stable control in real-world scenarios through adaptive calibration of simulation parameters (e.g., friction coefficient, actuator delay) based on real-time sensory feedback.

- **Environmental Drift:** Seasonal variations in lighting, weather, and background clutter can degrade the performance of perception models in long-term outdoor deployments, such as in agricultural or search-and-rescue scenarios [39].

Emerging research is tackling these issues. Diffusion-based domain randomization is being used to generate highly diverse and realistic synthetic training data with variable textures and illumination, significantly improving sim-to-real alignment and reducing deployment errors [14]. In addition, embodied intelligence (i.e., intelligence grounded in physical interaction with the environment) frameworks that integrate proprioceptive feedback with vision-language models are proving effective. By leveraging physical interaction data, these frameworks allow robots to self-calibrate and adapt their models in unstructured environments, thereby closing the sim-to-real gap more effectively than vision-only approaches [40]. Table 4 quantitatively compares the generalization capabilities and limitations of different adaptation frameworks.

Table 4. Cross-Domain Generalization Performance Comparison

No.	Method & Scenario	Cross-Domain Accuracy	Data Efficiency	Sim-to-Real Gap	Environmental Robustness
1	MAML-VLM (Novel object grasping)	High accuracy on novel objects	Few demonstrations (~5)	Noticeable performance degradation	Affected by visual domain shifts (e.g., texture)
2	SimCLR-VLM (Cross-crop recognition)	High accuracy across crops	Self-supervised (no labels)	N/A (Trained on real data)	Affected by seasonal variations
3	EWC+GPT (Sequential manufacturing tasks)	Superior accuracy vs. naive fine-tuning	Faster retraining	Moderate error rate	Affected by drastic task variations
4	Diffusion Domain Randomization (Agricultural transfer)	Greatly improved sim-real alignment	Relies on synthetic data	Minimal degradation	Robust to environmental changes

2.5 System Integration: Bridging the Gap to Real-World Deployment

Deploying Large-scale Models (LSMs) in real-world robotics requires a holistic co-design approach encompassing algorithms, hardware, and software infrastructure that addresses stringent constraints on computation, power, latency, and safety. Moving beyond isolated benchmarks to integrated systems—especially for closing the sim-to-real gap in multi-stage tasks—represents the final and most critical frontier for LSM-powered autonomy [38].

A primary bottleneck is the computational intensity of transformer-based models. To address this, advanced model compression techniques are essential. Post-training quantization (e.g., INT8 precision with tensor-aware calibration) and pruning (e.g., removing non-essential self-attention weights) achieve significant model size reduction while maintaining high accuracy on edge hardware like the NVIDIA Jetson platform. This enables complex models to run on resource-constrained devices, though often with a trade-off in functionality retention [23].

Beyond general-purpose hardware, specialized processors offer dramatic gains in efficiency. Neuromorphic co-processors like Intel's Loihi 2 leverage event-based spiking neural networks (SNNs) to achieve exceptional energy efficiency for specific tasks. For instance, Loihi 2 has demonstrated instantaneous inference for surgical robot scene understanding with a power consumption of only 92mW, substantially extending operational time compared to GPU-based systems in continuous procedures [41]. Similarly, photonic tensor processors represent an emerging technology, exploiting optical computing to achieve unparalleled throughput and energy efficiency (up to 23.4 TOPS/W) for matrix operations fundamental to LLM inference at the edge [42].

On the software side, optimized inference frameworks are critical. Runtimes like ONNX Runtime and NVIDIA TensorRT enhance performance through kernel fusion, operator optimization, and graph-level transformations. In industrial settings, these optimizations have been shown to significantly improve the real-time responsiveness of robotic

arms by efficiently managing computational graphs and leveraging hardware accelerators [43].

Standardization and safety are paramount for widespread adoption. The robotics community is increasingly adopting interoperability standards like ROS 2 with dedicated LSM interfaces for plug-and-play deployment. For ethical and functional safety, standards such as IEEE 7009-2021 are being integrated into medical and assistive robotics systems. These standards mandate explainability logs (e.g., visualizing diagnostic attention maps) and bias audits to ensure minimal demographic performance variance, which is critical for trust and verification [44].

Despite these advancements, two critical barriers hinder ubiquitous deployment:

- **Hardware Fragmentation:** The diversity of compute architectures (ARM, x86, GPU, NPU) across robotic platforms creates a significant standardization gap, complicating software development and deployment for large-scale industrial applications [45].

- **Energy Inefficiency:** Many embedded platforms still achieve sub-5 TOPS/W performance, which severely limits the deployment duration and capability of battery-powered mobile robots in field applications [23].

Emerging solutions are tackling these system-level challenges. Ultra-lightweight architectures, such as TinyRobot-Transformer, leverage neural architecture search (NAS) to achieve extreme compression, maintaining high accuracy on agricultural drones with a power draw of only 0.2W [46]. Although its peak compute performance is only ~0.2 TOPS, the architecture achieves an energy efficiency of ~8.7 TOPS/W through NAS optimization—striking a critical balance between computational capability and power consumption that is well-suited for the long-endurance requirements of agricultural drones. Beyond that, initiatives are focusing on creating unified middleware and compiler frameworks (e.g., Apache TVM) that can automatically compile and optimize LSM graphs for diverse hardware backends, mitigating the fragmentation issue [47].

Table 5 benchmarks key hardware deployment platforms against the efficiency and compatibility metrics essential for real-world LSM-robotics integration.

Table 5. Hardware Deployment Platform Benchmarking

No.	Platform	Peak Compute (TOPS)	Energy Eff. (TOPS/W)	INT8 Support	Typical Use Case	Key Limitation
1	NVIDIA Jetson AGX Orin	200	4.5	✓	Industrial arms, autonomous navigation	High cost, active cooling required
2	Intel Loihi 2 (Neuromorphic)	~0.9	~23.4	✗	Low-power perception, event-based processing	Limited programming model, specialized use
3	Raspberry Pi 4	~0.05	~0.3	✓ (Limited)	Education, lightweight HRI	Severely limited compute for LSMs
4	Google Edge TPU	4	2	✓	On-device inference, smart cameras	Fixed architecture, memory-bound
5	TinyRobot-Transformer (NAS)	~0.2	~8.7	✓	Agricultural drones, long-endurance missions	Task-specific, requires design effort

3 Challenges and Future Directions

Despite the substantial progress detailed in this review, the

seamless integration of Large-scale Models (LSMs) into robust, reliable, and ubiquitous robotic systems faces persistent multidisciplinary challenges. This section

synthesizes these hurdles across technical, operational, and ethical dimensions and outlines promising pathways for future research.

3.1 Technical Bottlenecks: Real-time Performance, Robustness, and Adaptation

The computational complexity of LSMs remains the primary impediment to their widespread deployment in real-time control loops. While quantization and pruning offer partial solutions, they often incur considerable performance loss, particularly for smaller models on ultra-low-power devices [23]. Achieving real-time inference times (<100 ms) for complex vision-language tasks on platforms like Raspberry Pi remains an open challenge. Future efforts must prioritize the co-development of ultra-efficient model architectures (e.g., via neural architecture search [47]) and specialized hardware accelerators (e.g., photonic processors, neuromorphic chips [48]) to deliver the necessary computational density and energy efficiency for long-term autonomy.

However, the robustness of LSM-driven systems under distribution shift—such as novel object types, extreme weather conditions, or adversarial perturbations—is not yet sufficient for safety-critical applications. Future research should focus on enhancing resilience through uncertainty quantification and self-supervised online adaptation. Techniques that allow robots to continuously calibrate their perception and action models based on incoming sensor data stream will be crucial for operating in unpredictable real-world environments [38, 49].

3.2 Standardization and Interoperability Gaps

Hardware and software fragmentation remains a major impediment to ecosystem unification. The lack of standardized interfaces between different LSM architectures (e.g., various transformer backbones), robotic middleware (e.g., ROS 2), and hardware platforms (ARM, x86, GPU) creates significant integration overhead and stifles reproducibility [44]. A critical future direction is the community-driven development of unified APIs and modular frameworks for LSM-robotics integration. Initiatives akin to ROS 2-LSM interfaces and standardized benchmarking suites will be essential to accelerate innovation and enable plug-and-play compatibility across different robotic platforms [50].

3.3 Ethical and Safety Governance

As LSMs grant robots greater autonomy, ensuring ethical and safe operation remains a critical challenge. Challenges include preventing models from executing harmful instructions, mitigating societal biases embedded in training data, and ensuring transparent decision-making [27, 43]. Future work must move beyond post-hoc filtering towards embedding safety directly into the model's planning and control cycles. This includes the advancement of real-time safety governance frameworks that integrate formal verification methods (e.g., using Control Barrier Functions [21]) with LLM-based reasoning to provide provable safety guarantees. Even so, establishing auditing trails and explainability protocols

mandated explicitly by standards like IEEE 7009-2021 will be non-negotiable for deployment in healthcare, public spaces, and other sensitive domains [29].

3.4 Toward Embodied AGI: Promising Research Frontiers

Looking beyond current challenges, several frontiers hold exceptional promise for the future of LSM-powered robotics:

- **Language-Guided Meta-Learning:** As a foundational capability for breaking the ‘data-hungry’ bottleneck in robotic skill adaptation, developing systems that can use natural language prompts not just for task execution but for few-shot learning of new skills—by parsing task logic from language to generate simulated training distributions—dramatically reducing the need for extensive retraining or demonstration [51].

- **Physics-Aware and Grounded Reasoning:** As a key breakthrough to address the ‘physical implausibility’ of data-driven LSMs, integrating causal reasoning and physical dynamics models into LSMs to overcome the current limitations of purely data-driven pattern matching, enabling more predictable and physically plausible robot behavior [52].

- **Multi-Agent Collaboration:** Building on single-robot LSM capabilities to expand into large-scale complex scenarios, scaling LSM-driven coordination to heterogeneous multi-robot systems (e.g., drone and ground robot teams) for complex tasks like distributed search and rescue, requiring advances in communication-efficient federated learning (to share task knowledge across robots) and emergent communication strategies [25].

- **Sustainable Deployment:** As a prerequisite for long-term real-world adoption of LSM-robotics systems, Research into ultra-low-power design and lifelong learning algorithms is essential to reduce the environmental footprint of training and deploying large models and to enable sustainable robotics operating over years and decades [53].

Addressing these challenges will require unprecedented collaboration across the fields of robotics, machine learning, hardware design, ethics, and human-computer interaction. The trajectory is clear: the future of intelligent robotics lies not in isolated models or algorithms, but in integrated, secure, and ethically aligned systems that amplify human capabilities.

4 Conclusion

This review has provided a comprehensive analysis of the convergence of Large-scale Models (LSMs) and robotics, moving beyond a bibliometric perspective to deliver a technical synthesis of advancements, performance, and persistent challenges across five core application domains: multimodal perception, semantic planning and control, human-robot interaction, cross-domain generalization, and efficient system integration.

A primary contribution of this work is its systematic, quantitative benchmarking of LSM-driven capabilities against traditional methods, drawing on empirical data to reveal critical trade-offs in accuracy, latency, and robustness. The review establishes that LSMs are not merely auxiliary tools but are fundamentally reshaping robotic autonomy. They serve as a unifying foundation that bridges three key advances: cognitive architectures (e.g., transformers) that connect symbolic reasoning with subsymbolic control, generative-

probabilistic fusion that enables end-to-end perception-action loops with uncertainty awareness, and embodied representation learning that aligns multimodal embeddings through physical interaction. Empirically, these integrations have demonstrated tangible benefits, from enhanced surgical precision and reduced industrial assembly errors to efficient inference on edge devices powered by neuromorphic and quantized systems.

However, this review also highlights key constraints that restrict current deployments. Foremost among these are the unresolved challenges in achieving real-time performance on resource-constrained platforms, ensuring robustness under environmental and domain shifts, and establishing scalable safety and ethical governance frameworks. The field also grapples with a pronounced standardization gap in hardware and software interfaces, which exacerbates development overhead and hinders reproducibility.

To fully realize the potential of LSM-robotics integration, future development must prioritize several pathways. First, research must focus on improving perceptual robustness and adaptation under real-world uncertainties, potentially through physics-aware simulation and online self-supervised learning. Second, implementing real-time ethical autonomy requires co-designing safety-critical frameworks that embed formal verification, such as control barrier functions, within the LSM decision-making process. Third, advancing few-shot adaptation through language-guided meta-learning will be essential for scalable generalization. Finally, the community must advocate for and develop standardized, plug-and-play interfaces to mitigate fragmentation and accelerate deployment.

In conclusion, while the integration of LSMs has undeniably propelled robotics into a new era of adaptability and intelligence, its translation from laboratories to widespread real-world adoption hinges on addressing these multifaceted challenges through concerted interdisciplinary efforts. The future of the field lies not in isolated algorithmic advances but in the holistic co-design of robust, efficient, and ethically aligned embodied AI systems.

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